



**Local AI for
Genealogists**
Hardware Guide — Can
My Computer Run Local
AI?
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Hardware Guide: Can My Computer Run Local AI?

The short answer: If your computer is less than 5 years old and has at least 16 GB of memory, try it before buying anything new.

How to Check Your Computer

On a Mac: Apple Menu (top-left) → About This Mac Look for: Chip and Memory

On Windows: Start Menu → Settings → System → About Look for: Processor and Installed RAM

Mac Compatibility Guide

Chip	Memory	Result	Recommended Models
	8 GB		Qwen3 8B

Chip	Memory	Result	Recommended Models
M1, M2, M3, M4		Light use only — small models	
M1, M2, M3, M4	16 GB	Good beginner setup	Qwen3 VL 8B
M1, M2, M3, M4	24–32 GB	Excellent	Qwen3 VL 8B or 32B
M1, M2, M3, M4	64 GB+	Professional workstation	Qwen3 VL 32B
Intel (any year)	Any	Works, but slow — limited	Qwen3 8B only

Apple Silicon chips (M1 through M4) are the gold standard for local AI on Mac. They have unified memory that both the CPU and GPU can use, which makes running large AI models significantly faster than Intel.

Windows Compatibility Guide

RAM	GPU	Result
16 GB	Recent NVIDIA (RTX 3000+)	Good
32 GB	NVIDIA RTX 4000+	Excellent
16 GB	No dedicated GPU (integrated)	Slow — limited
16 GB	Older NVIDIA (GTX series)	Works but limited
8 GB	Any	Not recommended

Windows users: A dedicated NVIDIA GPU dramatically improves performance. AMD GPUs also work, with slightly less optimization. If you have a gaming laptop or desktop from 2021 or later, you are likely in good shape.

Storage Requirements

Models are large files. Plan for this storage usage:

Model	Storage Needed
Qwen3 8B	~5 GB
Qwen3 VL 8B	~8 GB
Qwen3 VL 32B	~20 GB
Nomic Embed Text v1.5	~1 GB
LM Studio application	~400 MB
Starting total (8B + embed)	~10 GB

Model	Storage Needed
Full setup (32B + embed)	~22 GB

Check your available space: You need at least 20 GB free to get started. To check: Mac → About This Mac → Storage. Windows → Settings → Storage.

Memory Usage During a Session

When a model is running, it uses most of your available memory. This is normal — the model is loaded entirely into RAM.

Memory	What This Means
8 GB total	Other apps may slow down while AI is running
16 GB total	Comfortable for AI + a few other apps
24 GB+	No noticeable impact on other applications

Tip: You do not need to close all other apps. But if your computer feels sluggish while the AI is running, try closing unused browser tabs.

Who Should Buy New Hardware?

Wait before buying if: - Your current machine is less than 5 years old - You have not yet tried LM Studio on your current machine - Local AI is new to you

Consider upgrading if: - You have tried LM Studio and performance is too slow for your workflow - You have an Intel Mac older than 2020 - You have a Windows laptop with less than 16 GB RAM and no GPU - You do genealogy research as a profession or serious hobby

Best value options (as of mid-2026): - **Mac:** M4 MacBook Air, 24 GB memory — starts around \$1,099 (runs Qwen3 VL 8B beautifully; can handle VL 32B with patience) - **Mac:** M4 Mac mini, 24 GB — starts around \$799 (excellent value) - **Windows:** Any laptop with NVIDIA RTX 4070 or better + 32 GB RAM

Activity Monitor Check (Mac)

Before running LM Studio, open Activity Monitor (Applications → Utilities) and look at the Memory tab. If Memory Pressure is already red or yellow before you start, your machine may struggle with larger models.